[INCH-POUND]
A-A-52597
February 23, 1996
SUPERSEDING
MIL-V-10386E(ME)
20 December 1989

COMMERCIAL ITEM DESCRIPTION

VALVES, CHECK: SWING, CAST-IRON AND STEEL

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers requirements for cast-iron and steel swing check valves, hereinafter referred to as “valves”, designed for liquid or gaseous petroleum products.

2. CLASSIFICATION. Valves shall be of the following sizes and classes, as specified (see 7.2).

Size

2-inch [50.8 millimeters (mm)]	12-inch (305 mm)
3-inch (76.2 mm)	14-inch (356 mm)
4-inch (102 mm)	16-inch (406 mm)
6-inch (152 mm)	18-inch (457 mm)
8-inch (203 mm)	20-inch (508 mm)
10-inch (254 mm)	24-inch (610 mm)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Commands, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

FSC 4820

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Class	Material Code
125-pound [56.7 kilograms (kg)]	I - Cast Iron
150-pound (68.0 kg)	II - Steel
250-pound (113 kg)	
300-pound (136 kg)	
400-pound (181 kg)	
600-pound (272 kg)	

3. SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's material specifications. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 4.1).

3.1.1 Corrosion resistance. The valves shall be fabricated from materials that are inherently corrosion resistant or treated to provide protection against the various types of corrosion that may be encountered.

3.1.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.2 Design and construction. The valves shall be designed to perform as specified herein with liquid or gaseous petroleum products. Flowlines through the valves shall be streamlined. The 6, 8, and 12 inch (152, 203, and 305 mm) valves shall pass a brush-type pipeline cleaner conforming to A-A-52519. Valve port openings shall be full opening. The minimum bore for the inside diameter (ID) of each valve size shall be as shown in Table I.

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TABLE I. Minimum bore.

Nominal valve size	Minimum bore (ID)
inch (mm)	inch (mm)
2 (50.8)	2 (50.8)
3 (76.2)	3 (76.2)
4 (102)	4 (102)
6 (152)	6 (152)
8 (203)	8 (203)
10 (254)	10 (254)
12 (305)	12 (305)
14 (356)	13.25 (337)
16 (406)	15.25 (387)
18 (457)	17.25 (438)
20 (508)	19.25 (489)
24 (610)	23.25 (591)

3.2.1 Cast-iron valves. Cast-iron valves shall conform to MSS SP-71. Unless otherwise specified (see 7.2), the seating shall be bronze.

3.2.1.1 End flanges. End flanges shall be integral with the valve body. Flange facings and drilling shall conform to ANSI B16.1. The flanges shall be finished on the joint side in accordance with MSS SP-6.

3.2.1.2 Covers. Material for the cast-iron valve covers shall be in accordance with ASTM A126, grade B or C.

3.2.1.2.1 Cover bolts. Bolts for the covers shall conform to ASTM A307, grade B, and shall be of the stud, bolt-stud, bolt, or capscrew type. The bolts shall have heavy, unfinished, hexagonal heads. The bolts shall be wet assembled.

3.2.2 Steel valves. Steel valves shall be manufactured in conformance to API SPEC 6D, and shall be either cast or forged steel.

3.2.2.1 End connections. Valve ends shall be flanged, threaded, or beveled for welding (steel valves only), as specified (see 7.2). Threaded ends shall not be furnished on valves with end connections larger than 4 inch (102 mm).

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3.2.2.2 End flanges. End flanges shall be integral with the valve body. Flange facings and drilling shall conform to ANSI B16.5. The flanges shall be finished on the joint side in accordance with MSS SP-6.

3.2.2.3 Covers. Covers shall be castings, forgings, or steel plates. The covers shall be removable for access to the interior of the valves.

3.2.2.3.1 Cover bolts. Bolts and studs for the covers shall conform to ASTM A449. The bolts shall have heavy, unfinished, hexagonal heads. The bolts shall be wet assembled.

3.2.2.4 Seats. Unless otherwise specified (see 7.2), the valves shall be furnished with preformed, hydrocarbon, fuel resistant, O-ring clapper ring seats capable of withstanding continuous temperatures from -65 to +160 degrees Fahrenheit (°F) [-54 to +71 degrees Celsius (°C)].

3.2.2.5 Metal-to-metal seats. Unless otherwise specified (see 7.2), seating material of steel valves having metal-to-metal seats shall be 11.5 to 13.4 percent chromium steel.

3.2.2.6 O-ring clapper seals. When specified (see 7.2), the seal between clapper and body seat faces shall be an elastomeric ring, securely lodged in a dovetail groove, positioned in either the clapper or body seat face. The elastomeric ring shall be field removable and replaceable, and shall be chosen for suitability in the service specified. In the absence of any such specification, the elastomer shall be suitable for fuels containing aromatics.

3.2.2.7 Welding ends. Steel valve ends shall be beveled for welding in accordance with ANSI B16.25, figure 2 or 3, as applicable.

3.2.3 Cast-iron and steel valves.

3.2.3.1 Outside lever and weight. When specified (see 7.2), valves shall be equipped with an outside lever and weight. The hinge pin shall extend through a pressure-tight stuffing box in the side of the valve body. The lever shall be keyed and locked to the hinge pinch The weight shall be adjustable along the length of the lever. Set screws or jam nuts shall secure the weight to the lever. Unless otherwise specified (see 7.2), the lever and weight shall balance the disc.

3.2.3.2 Flow indicator. Each valve body shall have an arrow on both sides to indicate the direction of flow through the valve. These arrows may be cast or stamped in the metal, or attached by spot welding or brazing. The edges of the arrows shall be rounded.

3.2.3.3 End flange gaskets. Unless otherwise specified (see 7.2), end flange gaskets shall be fabricated from 0.062 inch (1.6 mm) thick material.

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3.2.3.4 End flange bolting. When specified (see 7.2), end flange bolting shall be furnished. Bolts shall have regular square, or heavy hexagonal heads, or may be of the bolt-stud type with dimensions conforming to ANSI B18.2.2. For cast-iron valves, the bolts, bolt-studs, or nuts shall conform to ASTM A307, grade B. For steel valves, the bolts or bolt-studs shall conform to ASTM A449.

3.2.3.5 Threaded ends. Valve ends shall be threaded with internal NPT threads conforming to API SPEC 5B.

3.2.3.6 End-to-end dimensions. End-to-end dimensions shall conform to ANSI B16.10.

3.3 Identification and marking. Identification and marking shall be permanent and legible, and shall include as a minimum, the manufacturer's identification code (CAGE), the contract number, the part or identifying number (PIN) (see 7.3), and the national stock number (NSN).

3.3.1 Body marking. Body marking shall conform to MSS SP-25.

3.4 Treatment and painting. Unless otherwise specified in contract (see 7.2), the parts of the valve normally painted shall be cleaned, treated, and painted in accordance with the contractor's commercial practice. When a color choice is standard practice, camouflage green is preferred.

4. REGULATORY REQUIREMENTS.

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and tests.

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID, and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

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7. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 Government documents. Copies of CID A-A-52519 may be obtained from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.1.2 Non-Government publications. Copies of ANSI B16.1, B16.5, B16.10, B16.25, and B18.22 may be obtained from the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036. Copies of API specifications 5B and 6D may be obtained from the American Petroleum Institute, Production Division, 1220 L Street, NW, Washington, DC 20005. Copies of ASTM A126, A307, and A449 may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Copies of MSS SP-6, SP-25, and SP71 may be obtained from the Manufacturing Standardization Society of the Valves and Fitting Industry, 127 Park Street NE, Vienna, VA 22180.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of DODISS to be cited in the solicitation.
c. Size and class of valve required.
d. Seating material required if other than bronze.
e. Type of valve ends required for steel valves.
f. Type of seats required for steel valves if other than specified.
g. Seating material required for metal-to-metal seats if other than specified.
h. When O-ring clapper seal is required.
i. When an outside lever and weight are required.
j. When the lever and weight are not required to balance the disc.
k. When end-flange gaskets shall be fabricated from material other than specified.
l. When end-flange bolting shall be furnished.
m. PIN, and quantity of valves required.
n. Finish, if other than as specified.
o. Applicable level of preservation, packing, and marking.

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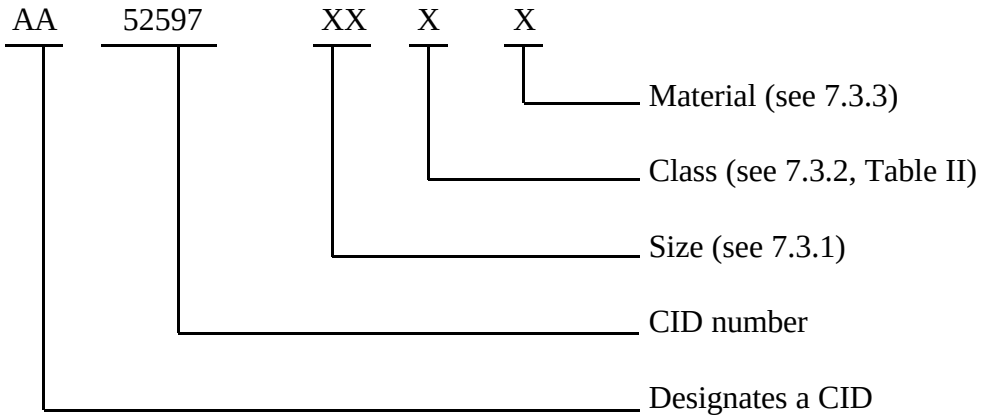
7.3 Part or identification number (PIN). The PINs to be used for valves acquired to this CID are created as follows:



7.3.1 Size code. The size code for the valve shall be equal to the size of the valve, which is designated in inches.

EXAMPLE:

Valve Size	Size Code
24 inch	24

7.3.2 Class code. See Table II.

TABLE II. Class code.

Class pound (kg)	Code
125 (56.7)	1
150 (68)	2
250 (113)	3
300 (136)	4
400 (181)	5
600 (272)	6

7.3.3 Material code. The part number (PIN) shall have material code “A” for cast-iron valves, or code “B” for steel valves.

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7.4 Cross-reference data. Valves conforming to this CID are interchangeable/substitutable with valves conforming to MIL-V-10386E(ME).

MILITARY INTERESTS:

Custodian:
Army - AT

Review Activities:
Army - CE1
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 4820-0706)

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